

# CDKN2B fusion-hotspot benchmark report

Study: msk\_impact\_50k\_2026

## Abstract

This report analyzes CDKN2B gene fusions from the msk\_impact\_50k\_2026 study, covering 12 fusion events. 2/12 fusions (16.7%) were in-frame. 2/12 fusions (16.7%) retained the PF12796 domain. Domain retention was tested with Fisher's exact test ( $p=0.0222222$ , statistically significant at  $\alpha=0.05$ ). No literature benchmark is configured for CDKN2B.

## Results summary

### Fusion partner frequency

Fusion-partner frequency was tabulated across 12 analyzed fusion events, identifying 7 distinct fusion partner genes. The most frequent partner was CDKN2A, observed in 6 events.

### Domain retention

PF12796 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 2/2 (100.0%) of in-frame fusions retained the domain,  $p=0.0222222$  (statistically significant at  $\alpha=0.05$ ). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.041958.

### Domain disruption

Domain-disruption analysis was skipped: CDKN2B has no disruption\_required\_domains configured; domain-disruption analysis was skipped.

### Cutpoint detection

Cutpoint detection scanned 10 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 53 aa (permutation-corrected  $p=0.762376$ , not statistically significant at  $\alpha=0.05$ ). This is -1 aa from the nearest configured domain boundary at 54 aa.

### Corroborating confidence statistics

Corroborating confidence statistics compared Domain\_retention\_flags groups "retained" ( $n=2$ ) and "not\_retained" ( $n=8$ ). For Frame\_status, retained: 2/2 (100.0%, 95% CI 34.2%-100.0%); not\_retained: 0/1 (0.0%, 95% CI 0.0%-79.3%). Welch's t-test comparing Tumor\_variant\_count between groups gave means of 11.5 vs 48.25,  $p=0.170234$  (not statistically significant at  $\alpha=0.05$ ).

### Composite evidence score

Composite evidence score reported: domain\_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption\_required\_domains not configured for this gene).

### Exon retention

Exon retention produced results for this run (1 row(s) in Exon\_retention); see the accompanying table.

## Joint partner

Joint partner reported: CDKN2B has no gene\_pair configured; joint-partner analysis was skipped.

# Tables

## composite\_score tables

composite\_evidence\_ranking

| Partner_gene | Event_count | Recurrence_score    | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                            | Composite_score     | Rank |
|--------------|-------------|---------------------|------------------------|-------------------------|--------------------------|----------------------------|----------------------------------------------------------------------------------|---------------------|------|
| CDKN2A       | 6           | 0.5                 | 0.13771847870814183    | None                    | 0.6721311475409837       | 0.2744647709419452         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.39086971112797186 | 1    |
| CNTRL        | 1           | 0.08333333333333333 | 0.13771847870814183    | None                    | 1.0                      | 0.2744647709419452         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2960951209640374  | 2    |
| MTAP         | 1           | 0.08333333333333333 | 0.13771847870814183    | None                    | 1.0                      | 0.2744647709419452         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2960951209640374  | 3    |
| SLC24A2      | 1           | 0.08333333333333333 | 0.13771847870814183    | None                    | 0.7540983606557377       | 0.2744647709419452         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.24998856358698826 | 4    |
| GAB2         | 1           | 0.08333333333333333 | 0.13771847870814183    | None                    | 0.6721311475409837       | 0.2744647709419452         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.23461971112797192 | 5    |
| ELAVL2       | 1           | 0.08333333333333333 | 0.13771847870814183    | None                    | None                     | 0.2744647709419452         | ['confidence_certainty', 'domain_retention', 'recurrence']                       | 0.13365553349419998 | 6    |
| CDKN2B-AS1   | 1           | 0.08333333333333333 | 0.13771847870814183    | None                    | 0.0                      | 0.2744647709419452         | ['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.10859512096403748 | 7    |

## cutpoint\_detection tables

cutpoint\_scan

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio | p_value             | neg_log10_p_value  |
|----------|------------------------|------------------|------------------------|------------------|------------|---------------------|--------------------|
| 38       | 0                      | 2                | 1                      | 7                | 0.0        | 1.0                 | -0.0               |
| 52       | 0                      | 2                | 3                      | 5                | 0.0        | 1.0                 | -0.0               |
| 53       | 2                      | 0                | 3                      | 5                | None       | 0.44444444444444453 | 0.3521825181113624 |
| 69       | 2                      | 0                | 4                      | 4                | None       | 0.46666666666666668 | 0.3309932190414243 |
| 73       | 2                      | 0                | 5                      | 3                | None       | 1.0                 | -0.0               |
| 75       | 2                      | 0                | 6                      | 2                | None       | 1.0                 | -0.0               |

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio | p_value | neg_log10_p_value |
|----------|------------------------|------------------|------------------------|------------------|------------|---------|-------------------|
| 113      | 2                      | 0                | 7                      | 1                | None       | 1.0     | -0.0              |

## domain\_retention tables

*frame\_domain\_contingency\_table*

|   |   |
|---|---|
| 2 | 0 |
| 0 | 8 |

*domain\_retention\_descriptives*

| Domain_key      | Domain_name                | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|-----------------|----------------------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| auto_key_domain | Ankyrin repeats (3 copies) | 10                      | 2                    | 5               | 3                | 0.34104938271604934                             |

## exon\_retention tables

*Exon\_retention*

| Exon | Retained_event_count | Total_event_count | Retained_fraction   |
|------|----------------------|-------------------|---------------------|
| 2    | 2                    | 12                | 0.16666666666666666 |

## frequency tables

*Partner\_gene\_counts*

| Partner_gene | Event_count |
|--------------|-------------|
| CDKN2A       | 6           |
| CDKN2B-AS1   | 1           |
| CNTRL        | 1           |
| ELAVL2       | 1           |
| GAB2         | 1           |
| MTAP         | 1           |
| SLC24A2      | 1           |

## Per-event results (results.tsv)

| event_id                         | sample_id                 | patient_id | fusion_name               | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction    | domain_is_truncated | retained_domains | lost_domains               | disrupted_domains          | source_annotation_text                                                                                                                     | source_site_2_effect_on_frame |
|----------------------------------|---------------------------|------------|---------------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------------|---------------------|------------------|----------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-003<br>5576-T01-I<br>M6-2  | P-0035<br>576-T0<br>1-IM6 |            | CDKN2A<br>-CDKN2<br>B     | CDKN2A       | unknown      | five_prime  | 22006179           | 2               | 75                          | False                  | disrupted     | 0.2716049<br>38271604<br>9  | True                |                  |                            | Ankyrin repeats (3 copies) | CDKN2B (NM_004936) - CDKN2A (NM_000077) Rearrangement: c.224:C DKN2B_c.153:CDKN2Adel   Protein Fusion: mid-exon {CDKN2B:CDKN2A}            | NA                            |
| EVT-P-001<br>0183-T02-I<br>M6-10 | P-0010<br>183-T0<br>2-IM6 |            | CDKN2B<br>-CDKN2<br>A     | CDKN2A       | unknown      | five_prime  | 22006197           | 2               | 69                          | False                  | disrupted     | 0.1975308<br>64197530<br>85 | True                |                  |                            | Ankyrin repeats (3 copies) | CDKN2B (NM_004936) - CDKN2A (NM_058195) rearrangement: c.206:CDKN2B_c.194-11:CDKN2A del   Protein Fusion: mid-exon {CDKN2B:CDKN2A}         | NA                            |
| EVT-P-006<br>0581-T01-I<br>M7-11 | P-0060<br>581-T0<br>1-IM7 |            | CDKN2B<br>-CDKN2<br>A     | CDKN2A       | out-of-frame | five_prime  | 22008168           | 1               | 52                          | True                   | lost          | 0.0                         | False               |                  | Ankyrin repeats (3 copies) |                            | CDKN2B (NM_004936) - CDKN2A (NM_000077) rearrangement: c.156+629:CDKN2B_c.458-143:CDKN2Adel   Protein Fusion: out of frame {CDKN2B:CDKN2A} | NA                            |
| EVT-P-004<br>5019-T02-I<br>M6-12 | P-0045<br>019-T0<br>2-IM6 |            | CDKN2B<br>-CDKN2<br>A     | CDKN2A       | unknown      | five_prime  | 22006065           | 2               | 113                         | False                  | disrupted     | 0.7407407<br>40740740<br>7  | True                |                  |                            | Ankyrin repeats (3 copies) | CDKN2B (NM_004936) - CDKN2A (NM_058195) rearrangement: c.338:CDKN2B_c.303:CDKN2Adel   Protein Fusion: mid-exon {CDKN2B:CDKN2A}             | NA                            |
| EVT-P-003<br>8263-T01-I<br>M6-13 | P-0038<br>263-T0<br>1-IM6 |            | CDKN2B<br>-CDKN2<br>A     | CDKN2A       | unknown      | five_prime  | 22007679           | 1               | 52                          | True                   | lost          | 0.0                         | False               |                  | Ankyrin repeats (3 copies) |                            | CDKN2B (NM_004936) - CDKN2A (NM_000077) Rearrangement: c.156+1118:CDKN2B_c.192:CDKN2Adel   Protein Fusion: mid-exon {CDKN2B:CDKN2A}        | NA                            |
| EVT-P-000<br>9347-T01-I<br>M5-34 | P-0009<br>347-T0<br>1-IM5 |            | CDKN2B<br>-CDKN2<br>B-AS1 | CDKN2B-AS1   | unknown      | five_prime  | 22006062           | 2               | 114                         | False                  | disrupted     | 0.7530864<br>19753086<br>4  | True                |                  |                            | Ankyrin repeats (3 copies) | CDKN2B (NM_004936) rearrangement : c.341_chr9:g.22015263inv   Protein fusion: mid-exon (CDKN2B-CDKN2BAS)                                   | NA                            |
| EVT-P-004<br>5616-T01-I<br>M6-36 | P-0045<br>616-T0<br>1-IM6 |            | CDKN2B<br>-SLC24A<br>2    | SLC24A2      | unknown      | five_prime  | 22008840           | 1               | 38                          | False                  | lost          | 0.0                         | False               |                  | Ankyrin repeats (3 copies) |                            | CDKN2B (NM_004936) - SLC24A2 (NM_020344) rearrangement: c.113:CDKN2B_c.931-51322:SLC24A2del   Protein Fusion: mid-exon {CDKN2B:SLC24A2}    | NA                            |

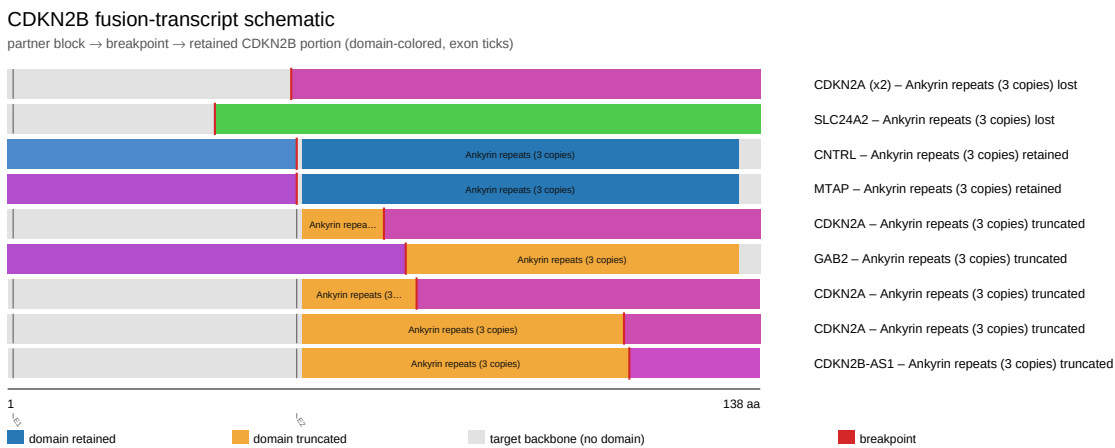
| event_id                         | sampl<br>e_id             | pa<br>tie<br>nt<br>_i<br>d | fusion_n<br>ame  | part<br>ner<br>_ge<br>ne | fra<br>me<br>_sta<br>tus | tar<br>get<br>_ro<br>le | breakp<br>oint_ge<br>nomic | break<br>point<br>_exo<br>n | breakpoint<br>_protein_p<br>osition | is_intron<br>ic_break<br>point | dom<br>ain_<br>statu<br>s | domain_r<br>etained_fr<br>action | domain<br>_is_trun<br>cated | retain<br>ed_do<br>mains                  | lost<br>_do<br>mai<br>ns | disrupt<br>ed_do<br>mains           | source_annotation_text                                                                                                                                                    | source_site<br>2_effect_on<br>_frame |
|----------------------------------|---------------------------|----------------------------|------------------|--------------------------|--------------------------|-------------------------|----------------------------|-----------------------------|-------------------------------------|--------------------------------|---------------------------|----------------------------------|-----------------------------|-------------------------------------------|--------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| EVT-P-004<br>2985-T01-I<br>M6-37 | P-0042<br>985-T0<br>1-IM6 |                            | CNTRL-<br>CDKN2B | CNT<br>RL                | in-fr<br>ame             | thre<br>e_<br>pr<br>ime | 220082<br>50               | 2                           | 53                                  | True                           | retai<br>ned              | 1.0                              | False                       | Ankyri<br>n repe<br>ats (3<br>copies<br>) |                          |                                     | CNTRL (NM_007018) -<br>CDKN2B (NM_004936)<br>rearrangement: c.1455+1<br>114:CNTRL_c.156+547:C<br>DKN2Binv   Protein<br>Fusion: in frame<br>{CNTRL:CDKN2B}                 | NA                                   |
| EVT-P-005<br>3040-T01-I<br>M6-38 | P-0053<br>040-T0<br>1-IM6 |                            | GAB2-C<br>DKN2B  | GAB<br>2                 | unk<br>now<br>n          | thre<br>e_<br>pr<br>ime | 220061<br>84               | 2                           | 73                                  | False                          | disru<br>pted             | 0.7654320<br>98765432<br>1       | True                        |                                           |                          | Ankyrin<br>repeats<br>(3<br>copies) | GAB2 (NM_080491) -<br>CDKN2B (NM_004936)<br>rearrangement: t(9;11)(p2<br>1.3;q14.1)(chr9:g.220061<br>84::chr11:g.78068634)  <br>Protein Fusion: mid-exon<br>{GAB2:CDKN2B} | NA                                   |
| EVT-P-000<br>2886-T01-I<br>M3-40 | P-0002<br>886-T0<br>1-IM3 |                            | MTAP-C<br>DKN2B  | MTA<br>P                 | in-fr<br>ame             | thre<br>e_<br>pr<br>ime | 220082<br>92               | 2                           | 53                                  | True                           | retai<br>ned              | 1.0                              | False                       | Ankyri<br>n repe<br>ats (3<br>copies<br>) |                          |                                     | NA   Protein fusion: in<br>frame (MTAP-CDKN2B)                                                                                                                            | NA                                   |

# Figures

## domain retention outliers



## fusion schematic



## reference comparison

Reference vs msk\_impact\_50k\_2026

